# Worksheet 5.1 — Limits and Introducing the Derivative

Mathematics: Analysis & Approaches SL (IB) · Unit 5: Differential Calculus

## The Big Ideas

Limits handle indeterminate forms; the derivative is defined as a limit of secant slopes.

- If direct substitution gives $\frac{0}{0}$, factor or rationalize first.
- $f'(x) = \lim_{h\to 0} \dfrac{f(x+h)-f(x)}{h}$.

## Worked Examples

### Example 1 — Factoring an indeterminate form

Evaluate $\lim_{x\to 5} \dfrac{x^2-25}{x-5}$.

***Solution.*** $\dfrac{(x-5)(x+5)}{x-5} = x+5 \to 10$.

**Conclusion:** 10.

### Example 2 — Rationalizing

Evaluate $\lim_{x\to 0} \dfrac{\sqrt{x+9}-3}{x}$.

***Solution.*** Multiply by the conjugate: $\dfrac{x}{x(\sqrt{x+9}+3)} = \dfrac{1}{\sqrt{x+9}+3} \to \dfrac{1}{6}$.

**Conclusion:** $\dfrac{1}{6}$.

### Example 3 — First principles

Use first principles to find the derivative of $f(x) = x^2 - 3x$.

***Solution.*** $f(x+h)-f(x) = 2xh + h^2 - 3h = h(2x+h-3)$; divide by $h$, let $h \to 0$.

**Conclusion:** $f'(x) = 2x - 3$.

### Example 4 — General quadratic from first principles

Use first principles to prove $\dfrac{d}{dx}[ax^2] = 2ax$.

***Solution.*** $f(x+h)-f(x) = a(x+h)^2 - ax^2 = a(2xh+h^2) = h(2ax+ah)$; divide by $h$, let $h \to 0$.

**Conclusion:** $f'(x) = 2ax$.

### Example 5 — Factoring a higher-degree indeterminate form

Evaluate $\lim_{x\to 3} \dfrac{x^3-27}{x-3}$ using the difference of cubes.

***Solution.*** $x^3-27=(x-3)(x^2+3x+9)$, so the limit simplifies to $x^2+3x+9 \to 9+9+9$.

**Conclusion:** 27.

### Example 6 — A limit requiring two factoring steps

Evaluate $\lim_{x\to 2} \dfrac{x^2-4}{x^2-5x+6}$.

***Solution.*** Factor both: $\dfrac{(x-2)(x+2)}{(x-2)(x-3)} = \dfrac{x+2}{x-3} \to \dfrac{4}{-1}$.

**Conclusion:** $-4$.

### Example 7 — First principles for a cubic

Use first principles to find the derivative of $f(x)=x^3$.

***Solution.*** $f(x+h)-f(x)=(x+h)^3-x^3=3x^2h+3xh^2+h^3=h(3x^2+3xh+h^2)$; divide by $h$, let $h\to 0$.

**Conclusion:** $f'(x)=3x^2$.

### Example 8 — Rationalizing with a linear numerator shift

Evaluate $\lim_{x\to 4} \dfrac{\sqrt{x}-2}{x-4}$.

***Solution.*** Multiply by the conjugate: $\dfrac{x-4}{(x-4)(\sqrt{x}+2)} = \dfrac{1}{\sqrt{x}+2} \to \dfrac{1}{4}$.

**Conclusion:** $\dfrac{1}{4}$.

### Example 9 — First principles at a specific point

Use first principles to find $f'(3)$ for $f(x)=x^2+1$ (evaluate the limit at the specific point $x=3$).

***Solution.*** $\dfrac{f(3+h)-f(3)}{h} = \dfrac{(3+h)^2+1-10}{h} = \dfrac{6h+h^2}{h} = 6+h \to 6$.

**Conclusion:** $f'(3)=6$.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Evaluate $\lim_{x\to 3}(2x^2 - x + 1)$.

### Question 2

Evaluate $\lim_{x\to -4}\dfrac{x^2 - 16}{x + 4}$.

### Question 3

Evaluate $\lim_{x\to 0}\dfrac{\sqrt{x+25} - 5}{x}$.

### Question 4

Use first principles to find the derivative of $f(x) = x^2 + 7x$.

### Question 5 — Challenge

Use first principles to find the derivative of $f(x) = 4x^2 - 2x$.

## Question 6 — Challenge

Evaluate $\lim_{x\to 2} \dfrac{x^3-8}{x-2}$ by factoring the numerator as a difference of cubes.

## Question 7

Evaluate $\lim_{x\to 3} \dfrac{x^2-9}{x^2-2x-3}$.

## Question 8 — Challenge

Use first principles to find the derivative of $f(x) = 2x^3$.

## Question 9

Evaluate $\lim_{x\to 9} \dfrac{\sqrt{x}-3}{x-9}$.

## Question 10

Use first principles to find $f'(2)$ for $f(x) = x^2 - 1$.

## Question 11

Evaluate $\lim_{x \to 1} \dfrac{x^3 - 1}{x - 1}$ using the difference of cubes.

## Question 12 — Challenge

Evaluate $\lim_{x \to -2} \dfrac{x^2 - 4}{x^2 + 3x + 2}$.

## Question 13 — Challenge

Use first principles to find the derivative of $f(x) = x^2 + 5x - 2$ (note the constant term's effect).

# Question Recap & Answer Key

## Question Recap

1. Evaluate $\lim_{x\to 3}(2x^2 - x + 1)$.
2. Evaluate $\lim_{x\to -4}\dfrac{x^2-16}{x+4}$.
3. Evaluate $\lim_{x\to 0}\dfrac{\sqrt{x+25}-5}{x}$.
4. Use first principles to find the derivative of $f(x) = x^2 + 7x$.
5. Use first principles to find the derivative of $f(x) = 4x^2 - 2x$.
6. Evaluate $\lim_{x\to 2}\dfrac{x^3-8}{x-2}$ by factoring the numerator as a difference of cubes.
7. Evaluate $\lim_{x\to 3}\dfrac{x^2-9}{x^2-2x-3}$.
8. Use first principles to find the derivative of $f(x) = 2x^3$.
9. Evaluate $\lim_{x\to 9}\dfrac{\sqrt{x}-3}{x-9}$.
10. Use first principles to find $f'(2)$ for $f(x) = x^2 - 1$.
11. Evaluate $\lim_{x\to 1}\dfrac{x^3-1}{x-1}$ using the difference of cubes.
12. Evaluate $\lim_{x\to -2}\dfrac{x^2-4}{x^2+3x+2}$.
13. Use first principles to find the derivative of $f(x) = x^2 + 5x - 2$ (note the constant term's effect).

## ✓ Answer Key

1. 16
2. $x - 4 \to -8$
3. $\dfrac{1}{\sqrt{x+25}+5} \to \dfrac{1}{10}$
4. $f'(x) = 2x + 7$
5. $f'(x) = 8x - 2$
6. $(x-2)(x^2+2x+4)/(x-2) = x^2+2x+4 \to 4+4+4 = 12$
7. $\dfrac{(x-3)(x+3)}{(x-3)(x+1)} = \dfrac{x+3}{x+1} \to \dfrac{6}{4} = \dfrac{3}{2}$
8. $f(x+h) - f(x) = 2(3x^2h + 3xh^2 + h^3) = h(6x^2 + 6xh + 2h^2)$; $f'(x) = 6x^2$
9. $\dfrac{1}{\sqrt{x}+3} \to \dfrac{1}{6}$
10. $\dfrac{(2+h)^2 - 1 - 3}{h} = \dfrac{4h + h^2}{h} = 4 + h \to 4$
11. $(x-1)(x^2+x+1)/(x-1) = x^2+x+1 \to 1+1+1 = 3$
12. $\dfrac{(x-2)(x+2)}{(x+1)(x+2)} = \dfrac{x-2}{x+1} \to \dfrac{-4}{-1} = 4$
13. $f(x+h) - f(x) = 2xh + h^2 + 5h = h(2x + h + 5)$; $f'(x) = 2x + 5$ (the constant $-2$ contributes nothing, as expected)